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Assessment of AI ethical reflection: the development and validation of the AI ethical reflection scale (AIERS) for university students

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Abstract

This study developed and validated the AI Ethical Reflection Scale (AIERS), a tool for measuring university students' ethical reflection on AI in three dimensions: AI ethical awareness, AI critical evaluation, and AI for social good. This study used a sample of 730 university students, and confirmatory factor analysis supported the three-dimensional structure of the AIERS, which demonstrated good internal consistency and construct validity. Additionally, we found evidence of convergent and discriminant validity, as AI ethical reflection was positively yet modestly correlated with AI literacy, indicating that AI ethical reflection is a related yet distinct construct from AI literacy. The study also explored the differences in AI ethical reflection based on gender, academic discipline, and prior AI experience. The results revealed that female students exhibited higher AI ethical awareness than male students. Furthermore, frequent AI users reported significantly greater AI ethical reflection across all dimensions than those who used AI less frequently or had never used it. However, no significant differences were found between Science, Technology, Engineering, Mathematics (STEM) and non-STEM students. Collectively, these findings emphasize the importance of integrating ethical considerations and hands-on AI experiences from diverse perspectives of gender in AI education across disciplines to promote the ethical and responsible use of AI among university students.

Keywords AI ethics, Ethical reflection, University students

Artificial intelligence (AI) has become ubiquitous in nearly all professions. Especially with the introduction of ChatGPT, a generative AI (GenAI) tool based on large language models, the general public has increasingly easy access to AI technologies. ChatGPT and other GenAI tools, along with other voice and image recognition, have transformed our daily lives. To empower people to adapt to the opportunities and challenges presented by AI, AI education has emerged as a global initiative (Singh et al., 2023). University students, the next generation of the workforce, have become the primary target of AI education (Chai et al., 2024). To better prepare students in all fields for a labor market

profoundly affected by AI, many higher education institutions have begun integrating AI into their curriculums (Southworth et al., 2023).

Although AI is bringing revolutionary changes, AI technologies can be misused, resulting in negative consequences for both the individual and society (Federspiel et al., 2023; Paraman & Anamalah, 2023). For example, deepfakes, which are hyper-realistic images, videos, or audio clips generated by AI to impersonate someone and present them doing or saying things that they never actually did, are increasingly being used for criminal purposes (Westerlund, 2019). Moreover, AI outputs are influenced by the data the AI is trained on, and discriminatory and harmful outcomes can result when biased data are used (Ray, 2023). All of these issues underscore the urgent need for a greater emphasis on AI ethics. In response, researchers have called for the integration of ethics into AI education and the creation of valid assessments to evaluate individuals' capacity for ethical reflection in the context of AI (Goldsmith et al., 2020).

Growing academic efforts are being made to analyze and assess essential capabilities that enable individuals to effectively understand, evaluate, and communicate with AI (e.g., Long & Magerko, 2020; Wang et al., 2023). However, studies and scales have been criticized for primarily focusing on individuals' basic knowledge of AI and skills in their applications (e.g., Laupichler et al., 2023; Pinski & Benlian, 2023). Although these scales often include dimensions related to ethical reflection, they largely address individuals' general AI ethical awareness. Nonetheless, it has been argued that AI literacy in ethics can extend beyond awareness and can involve behavioral tendencies that reflect individuals' critical evaluation of AI and its responsible use for the benefit of society (Chiu et al., 2024; Chiu, Chen, Chiu et al., 2024a, b). Therefore, this study sought to provide a more specific and comprehensive framework for understanding and measuring university students' ethical reflection on AI and to develop the AI Ethical Reflection Scale (AIERS). By identifying the key factors that shape individuals' AI ethical reflection, this study aimed to offer insights to enhance AI education, enabling individuals to acquire the vital skills needed to critically evaluate and engage ethically with AI.

AI ethical reflection and AI education

With the rapid advancements and widespread applications of AI technologies, individuals can benefit from AI in various domains, such as healthcare, entertainment, education, and transportation. However, the misuse of AI is also increasing, which can lead to serious ethical issues and exert significant detrimental effects on both the individual and society (Federspiel et al., 2023; Paraman & Anamalah, 2023). Given these ethical challenges, the frameworks for AI education must evolve accordingly. Although AI education is being increasingly promoted and implemented globally to prepare higher education students to adapt to an AI-based society, the primary focus of these educational programs has been to teach technological knowledge and skills concerning how to use AI for work or academic purposes (e.g., Zhai et al., 2021). There has been a comparative lack of education on AI ethics. Accordingly, researchers have stressed that AI education should go beyond literacy and foster ethical reflection (Goldsmith et al., 2020; Jobin et al., 2019).

To ensure effective and ethical engagement with AI, individuals must not only acquire the basic knowledge to comprehend how AI operates but also develop the competencies

to reflect on the ethical issues related to the application of AI, critically evaluate its outputs, and use it responsibly for social good (Ng et al., 2021). However, the emphasis on AI ethics has predominantly been directed toward stakeholders in engineering-related disciplines (e.g., Borenstein & Howard, 2021; Tomašev et al., 2020), with little attention given to ethical reflection competence among a more general population. Given that university students have increasingly become a target of AI education (Chai et al., 2024), this study extended research on AI ethical reflection to this group. Our main purpose was to contextualize the construct of AI ethical reflection to identify key components in the development of ethical competence related to AI among non-AI specialization university students, thereby informing the design of more tailored ethics education within the AI curriculum.

The components of AI ethical reflection

Individuals' capacity for ethical reflection on AI has received increasing attention in the literature. For example, Goldsmith et al. (2020) argued that AI ethical reflection can consist of two primary facets: ethical description and felt ethical responsibility. *Ethical description* involves individuals' ability to recognize and articulate ethical standards and issues in the development and usage of AI. It encompasses a general awareness of how individuals, corporations, and governments should use AI responsibly. *Felt ethical responsibility*, in contrast, refers to individuals' commitment to interacting with AI in an ethical manner. It reflects the motivation to ensure that the application of AI contributes positively to society.

Although these two facets can serve as a valuable theoretical foundation for understanding AI ethical reflection, researchers have not yet offered a thorough framework to fully conceptualize the scope of these components. To bridge this gap, the current study used the model of moral functioning developed by Rest (1986) to identify the underlying factors that shape individuals' AI ethical reflection. Rest's model has long been a critical framework used in the study of ethical decision-making competence across various domains (e.g., DeTienne et al., 2021; Pijanowski, 2017). The model outlines a structured process regarding how individuals reflect on ethical issues and make ethical decisions. It hypothesizes four key stages of ethical reasoning and behaviors: moral sensitivity, judgment, motivation, and implementation.

Moral sensitivity refers to the recognition of ethical problems. Moral judgment involves the process of evaluating whether a course of action is ethically appropriate. These two components closely align with the facet of ethical description proposed by Goldsmith et al. (2020), as both involve perceiving and critically assessing ethical considerations. Moral motivation entails the prioritization of ethical principles over competing interests. Moral implementation encompasses taking practical actions based on ethical decisions. These two components directly correspond to the facet of felt ethical responsibility, and both emphasize the personal drive to act on ethical principles and assume a positive social role. Despite the alignment between Rest's model and these two facets, applying this framework specifically to AI ethics requires further contextualization. Therefore, this study built on the two facets of ethical description and felt ethical responsibility of Goldsmith et al. (2020) and Rest's (1986) model of morality to contextualize AI ethical reflection in three dimensions: AI ethical awareness, AI critical evaluation, and

AI for social good. A detailed elaboration of the relevance of these underlying factors to AI ethical reflection is provided in the following sections.

AI ethical awareness

Individuals' behavioral intentions are fundamentally influenced by their knowledge bases and belief systems, which function as crucial cognitive precursors that inform and shape their decision-making processes (Fishbein & Ajzen, 2009). A general awareness of AI ethics can be a prerequisite for responsible AI use (Tigard, 2021). This dimension reflects Rest's (1986) concept of moral sensitivity, as it focuses on recognizing and articulating the ethical implications of AI. Previous research on AI ethics has predominantly examined this element as the representation of AI ethical reflection, underscoring AI ethical awareness as the ability to discern ethical issues in AI applications and understand how various stakeholders, including individuals, corporations, and governments, should responsibly manage AI (e.g., Hagendorff, 2020; Jobin et al., 2019). There is an increasing consensus that AI ethical awareness consists of five major aspects: transparency, fairness, privacy, responsibility, and non-maleficence (Jang et al., 2022; Jobin et al., 2019).

Transparency refers to the importance of AI systems' capacity to provide clear explanations of their technical procedures. Fairness relates to the salience of mitigating prejudice and discrimination for the development and application of AI systems that are inclusive and representative of diverse populations (European Commission, 2019; Li et al., 2024). Privacy pertains to the need for strict regulations to protect users' privacy throughout the lifecycle of AI systems (European Commission, 2019). Responsibility encompasses understanding the necessity of establishing mechanisms for accountability to ensure stakeholders' responsibility for the outcomes of AI systems (European Commission, 2019). Non-maleficence involves advocating for developing, deploying, and utilizing AI systems in a manner that avoids potential harm to human beings (Floridi et al., 2018). By acquiring this general AI ethical awareness, individuals can better anticipate the potential risks of AI and can have more positive and ethical engagement with AI (Jang et al., 2022).

AI critical evaluation

Many AI systems work as 'black boxes,' which makes it difficult for users to learn how their outputs are generated, potentially posing ethical concerns. This lack of transparency highlights the need for individuals to cultivate competencies in critical evaluation that can empower them to evaluate the legitimacy and potential biases of AI outputs and determine whether these outputs adhere to general ethical standards (Von Eschenbach, 2021). In alignment with Rest's (1986) model, which identifies moral judgment as an essential process of ethical decision-making, AI critical evaluation can be considered a vital element of AI ethical reflection. It pertains to individuals' ability to critically evaluate the information provided by AI and to make judgments about the appropriateness of the outputs of AI (Tian et al., [accepted](#)).

Prior literature has suggested that critical thinking is a key factor related to the process of moral judgment (e.g., Samanci, 2015). Critical thinking involves the ability to apply logical and reflective analysis to judge what to believe or to do (Lai, 2011). While AI is a powerful tool, it can be affected by algorithmic bias. When individuals follow potentially

prejudiced or even harmful recommendations from AI, this uncritical trust may lead to adverse outcomes such as discrimination and social exclusion (Araujo et al., 2020; Kapania et al. 2022; Zajko, 2021). To reduce this risk, critical thinking may be imperative in AI ethical reflection, which represents individuals' capability to critically reflect on the processes and outputs of AI to identify and address potential biases inherent in AI systems.

AI for social good

To capture the facet of felt ethical responsibility, this study further conceptualized the third dimension of AI ethical reflection, AI for social good. Recognized as a pivotal principle of ethical AI (Akula & Garibay, 2021; Floridi et al., 2018), AI for social good encompasses the motivation and behavioral tendency to ethically use AI in a manner that promotes positive social outcomes. Corresponding to Rest's (1986) theory that ethical development involves not only recognizing and evaluating moral issues but also motivating individuals to act in accordance with these moral principles, AI for social good reflects the final two components of moral motivation and implementation, emphasizing prioritizing and actualizing ethical considerations. It embodies individuals' commitment to ensuring that AI plays a socially beneficial role rather than aggravating inequalities or causing harm (Floridi et al., 2018). By focusing on AI for social good in the framework of AI ethical reflection, this dimension allows for an assessment of individuals' awareness and competence in transforming ethical principles into AI practice. This ethical reflection can be related to a sense of internalized responsibility for the ethical application of AI, which is particularly salient in light of the growing risks associated with AI (Paraman & Anamalah, 2023).

The current study

As AI has permeated almost all aspects of our daily lives, understanding and fostering ethical reflection on AI is vital to ensure that AI systems are developed and applied ethically and responsibly, thus avoiding potential risks associated with the inappropriate use of AI. In response to the growing ethical challenges accompanied by the applications of AI, there has been a burgeoning emphasis on integrating ethical education into AI curriculums, especially for university students, who are at the forefront of AI education (Alam, 2023). However, scarce research has provided a comprehensive model to operationalize and measure students' ethical considerations and competencies in engagement with AI. Therefore, this study aimed to address this gap by developing and validating an instrument to measure multiple dimensions of AI ethical reflection among university students. Drawing on Rest's (1986) model of morality and the two facets of ethical description and felt ethical responsibility proposed by Goldsmith et al. (2020), this study hypothesized that three dimensions – AI ethical awareness, AI critical evaluation, and AI for social good – capture the essential facets of ethical reflection in the context of AI. Through confirmatory factor analyses, this study sought to demonstrate the three-dimensional structure of the AIERS and validate its reliability and validity. Additionally, we examined whether and how AI ethical reflection would vary across student populations based on gender, prior AI experience, and academic discipline. Existing literature has suggested that moral development can differ by gender because of distinct gender-related socialization patterns (Gilligan, 1982; You et al., 2011), which warrants investigation of potential gender differences in AI ethical reflection. Furthermore, previous

research has documented significant differences in attitudes toward AI ethics between students who have previous experience with AI education and those who do not (Jang et al., 2022). It has been hypothesized that students' backgrounds play a significant role in how they reflect on the ethical dimensions of AI. Therefore, the current study also explored whether there would be disparities in AI ethical reflection among students according to their academic discipline (Science, Technology, Engineering, Mathematics [STEM] vs. non-STEM) and prior AI use. With the AIERS, educators and policymakers can better evaluate university students' capacity for ethical reflection on AI. Moreover, the assessment of AI ethical reflection across different demographical factors can guide the development of targeted AI curriculums to promote this necessary competence among diverse student populations, ultimately achieving more socially beneficial outcomes.

Method

Participants and procedure

The study received approval from the University Ethics Committee. Data collection was conducted through an online questionnaire distributed during the 2023–2024 academic year. The questionnaire consisted of two sections: the first asked for the participants' demographic information, such as university, discipline, age, gender, and prior AI experiences, and the second focused on the core study variables. Before proceeding with the questionnaire, the students were informed about the voluntary and autonomous nature of their participation and provided their consent. A total of 730 students from two universities in mainland China and Hong Kong completed the survey. The majority of these participants were female students (60.7%), with ages ranging from 17 to 30 years ($M = 21.23$, $SD = 2.84$). Regarding academic disciplines, only 46.4% of the participants were from STEM fields, and the remaining 53.6% were from humanities and social sciences, including social science, education, arts, business, and law. Regarding previous AI use experiences, the majority of students (64.0%) reported 'less frequent' use of AI for learning and assessment during the last term, followed by 27.8% who indicated 'frequent' use and 8.2% who reported having 'never' used AI.

Item generation

To maintain conceptual clarity and formulate the items for the three subcomponents of AI ethical awareness, AI critical evaluation, and AI for social good that we hypothesized would constitute the factor structure of AI ethical reflection, we examined the existing literature and scales related to AI ethics and moral development (e.g., Chai et al., 2020; Jang et al., 2022; Tian et al., *accepted*). Most of the items in the AIERS were adopted or adapted from previously validated scales (see Table 1). To ensure content validity while considering the practical constraints of a lengthy survey, we carefully selected the most representative items to capture the essence of each dimension. Each item included in the scale was further reviewed by three experts in the field of higher education to verify their appropriateness for the university student population. Finally, 12 items were established for the AIERS, and each item was measured using a 7-point Likert scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). A 7-point Likert scale provides sufficient points of differentiation without overcomplicating the response options. It can offer participants more opportunities to reflect on their true perspectives while avoiding cognitive

Table 1 AIERS items

Dimensions	Items	Adapted from
AI ethical awareness	1. When creating an AI system designed to suggest personalized clothing choices, I would recommend integrating data from individuals representing diverse body types. 2. I think it is essential for AI to explain the reasons for its decisions. 3. I think a lot of people should be educated on relevant ethical issues to avoid the potential misuse of AI. 4. I think when developing AI with people's personal information, caution must be exercised to avoid infringing on their privacy. 5. I think there needs to be a clear allocation of responsibilities within the organization using AI, and the creation of potential scenarios and ways to deal with harm when they occur.	Jang et al. (2022)
AI critical evaluation	1. I can think of other possible solutions besides those provided by AI. 2. I can critically evaluate the credibility of information provided by AI. 3. I would attempt to understand the underlying rationale of why AI provides particular answers. 4. I would compare what AI says with what I know.	Self-constructed
AI for social good	1. I would use AI to serve others. 2. I would use AI to help disadvantaged people. 3. I will take part in AI design projects that help others.	Chai et al. (2020)

overload, which can reduce central tendency bias and increase measurement reliability and validity (Joshi et al., 2015; Taherdoost, 2019).

Validity indicator and measure

To test the validity of the AIERS, we selected AI literacy as the convergent and discriminant validity measure. AI literacy reflects individuals' understanding of AI technologies and their applications, with ethical considerations being a crucial dimension (Chiu et al., 2024a; Ng et al., 2021). Individuals with higher AI literacy may be more capable of critically evaluating AI and recognizing its ethical implications. Consequently, it was expected that higher AI literacy would be significantly and positively associated with greater AI ethical reflection. In this study, five items from Laupichler et al.'s (2023) scale for the assessment of non-experts' AI literacy were used to evaluate the students' AI literacy. A sample item is 'I can critically reflect on the potential impact of AI on individuals and society'. The items were rated on a 7-point Likert scale, ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). Cronbach's alpha for the present study was 0.77, suggesting good measure reliability.

Data analysis

The data analysis for this study was undertaken using SPSS 26.0 and AMOS 26.0. To ensure the appropriateness of the subsequent model estimations, prior to the main analyses, the data were first tested for normality with skewness and kurtosis values. Next, the suitability of the data for factor analysis was assessed via the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity. Then, a series of confirmatory factor analyses (CFAs) were conducted to test the hypothesized factor structure of the AIERS. Specifically, one-factor and three-factor models were examined sequentially to determine which best represented the structure of AI ethical reflection. We first evaluated a one-factor model (Model 1), where all of the items were loaded onto a single general factor of AI ethical reflection. Subsequently, Model 2, a three-factor correlated model, was assessed, which assumed that AI ethical reflection comprises three distinct but interrelated dimensions (i.e., AI ethical awareness, AI critical evaluation, and

AI for social good). The fit of each model was evaluated and compared using the ratio of χ^2 to degrees of freedom (χ^2/df), goodness-of-fit index (GFI), comparative fit index (CFI), Tucker–Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR). In line with the established criteria, a model was considered to have an acceptable fit if the χ^2/df ratio was below 5 (Hair et al., 2014), the GFI, CFI, and TLI values exceeded 0.90 (Bentler, 1990), and the RMSEA and SRMR values were less than 0.08 (Byrne, 2010; Hu & Bentler, 1999). After the final model was established, we performed invariance testing to explore whether the model showed measurement invariance across male and female students. Specifically, the model was progressively constrained to evaluate four levels of invariance using χ^2 difference tests: configural invariance (no constraints across genders), metric invariance (equal factor loadings across genders), scalar invariance (equal latent factor covariances and variances across genders), and strict invariance (equal measurement error variances across genders). Additionally, potential differences in AI ethical reflection across demographic variables (i.e., gender, academic discipline, and previous AI experience with learning) were further examined through independent *t*-tests and analysis of variance (ANOVA).

Results

Preliminary analyses

All of the measured items demonstrated acceptable skewness values (ranging from -1.48 to -0.44) and kurtosis values (ranging from -0.03 to 2.62), which were below the maximum threshold of $|3|$, suggesting that the data for all of the items were approximately normally distributed (Farooq, 2016).

Validity of the AIERS

To ensure that our data were suitable for factor analysis, both Bartlett's test of sphericity and the KMO test were conducted. The results revealed that the result for Bartlett's test of sphericity was significant [$\chi^2 (66) = 2,694.20, p < .001$], and the KMO value was 0.85 , greater than 0.70 , indicating that the data were well suited for factor analysis (Hutcheson & Sofroniou, 1999; Pallant, 2016). Moreover, as shown in Table 2, all 12 items had factor loadings above the recommended threshold value of 0.40 , ranging from 0.47 to 0.81 , demonstrating that they adequately represented their respective constructs (Byrne, 2010). Furthermore, all three dimensions exhibited satisfactory internal consistency, with Cronbach's alpha coefficients ranging from 0.73 to 0.81 , exceeding the cutoff value of 0.70 (Fornell & Larcker, 1981). The composite reliability coefficients ranged from 0.74 to 0.81 , all higher than the recommended benchmark of 0.70 , and the average variance extracted (AVE) values ranged from 0.41 to 0.59 , greater than the minimum threshold of 0.40 , providing further support for the good internal consistency and convergent validity of the constructs in the AIERS (Fornell & Larcker, 1981).

To confirm the factor structure of AI ethical reflection, we further tested and compared the two models. Model 1 demonstrated poor fit: $\chi^2 (54) = 939.50$, $\chi^2/\text{df} = 17.40$, $\text{GFI} = 0.79$, $\text{CFI} = 0.67$, $\text{TLI} = 0.59$, $\text{RMSEA} = 0.15$, and $\text{SRMR} = 0.10$. These indices indicate that the one-factor model did not adequately represent the data. In contrast, Model 2 showed a significantly improved and identical fit to the data: $\chi^2 (51) = 216.76$, $\chi^2/\text{df} = 4.25$, $\text{GFI} = 0.95$, $\text{CFI} = 0.94$, $\text{TLI} = 0.92$, $\text{RMSEA} = 0.07$, and $\text{SRMR} = 0.05$. The

Table 2 CFA analysis of the questionnaire responses

Constructs and items	t-value	Factor loading	AVE	CR	Cronbach's alpha
AI ethical awareness (AIEA), $M=5.72$, $SD=0.80$			0.41	0.78	0.77
AIEA1		0.58			
AIEA2	10.03***	0.47			
AIEA3	12.82***	0.67			
AIEA4	13.46***	0.74			
AIEA5	13.32***	0.72			
AI critical evaluation (AICE), $M=5.23$, $SD=0.84$			0.42	0.74	0.73
AICE1		0.64			
AICE2	13.23***	0.77			
AICE3	11.42***	0.55			
AICE4	12.33***	0.61			
AI for social good (AISG), $M=4.87$, $SD=1.10$			0.59	0.81	0.81
AISG1		0.73			
AISG2	17.22***	0.81			
AISG3	17.16***	0.76			

Note. CR=composite reliability; AVE=average variance extracted

*** $p < .001$

Table 3 Correlations between the factors and AVE of the factors

	AI for social good	AI critical evaluation	AI ethical awareness
AI for social good	.77		
AI critical evaluation	0.51	0.65	
AI ethical awareness	0.38	0.61	0.64

Note. The bold numbers on the diagonal are the square root values of the average variance extracted (AVE) of each factor. The non-diagonal absolute value is the correlation coefficient of each factor. All of the correlations are significant ($p < .001$)

significant improvement in model fit compared with Model 1 ($\Delta\chi^2 = 722.74$, $\Delta df = 3$, $p < .001$) suggests that Model 2 better represented the data. In addition, Pearson's correlation coefficients of the dimensions were computed to assess the relationships between them, and all of them were significant (see Table 3). Furthermore, the square root of the AVE for each dimension exceeded the correlations between that dimension and the other dimensions, providing further evidence of good discriminant validity.

Invariance testing was conducted to verify whether the final model demonstrated equivalence across genders. Four models were evaluated progressively. The unconstrained model (configural invariance), where no constraints were applied, allowing the parameters to be estimated separately for male and female students, was compared with three increasingly restrictive models: the measurement weight model (metric invariance), where factor loadings were constrained to be equal across genders; the structural covariance model (scalar invariance), where both latent factor covariances and variances were constrained to be equal across genders; and the measurement error model (strict variance), where error variances were further constrained to be equal across genders. The results indicated that invariance was achieved for both the measurement weight model ($\Delta\chi^2 = 9.50$, $\Delta df = 9$, $p > .05$) and structural covariance model ($\Delta\chi^2 = 14.97$, $\Delta df = 15$, $p > .05$). In contrast, the measurement error model was significant compared with the unconstrained model, $\Delta\chi^2 = 51.08$, $\Delta df = 27$, $p = .003$, suggesting non-invariance. Collectively, these findings confirm the measurement invariance of the AIERS across genders at the levels of measurement weights and structural covariances.

Table 4 Correlations between the factors and AVE of the factors

	1	2	3	4	5
1. AI ethical awareness					
2. AI critical evaluation	0.47***				
3. AI for social good	0.32***	0.40***			
4. AI ethical reflection	0.80***	0.79***	0.72***		
5. AI literacy	0.19***	0.44***	0.39***	0.43***	
<i>M</i>	5.72	5.23	4.87	5.34	4.78
<i>SD</i>	0.80	0.84	1.10	0.68	1.06

Note. $n = 730$

*** $p < .001$

Table 5 Results of independent T-Tests by gender

	<i>t</i>	<i>df</i>	<i>p</i>	Female students		Male students	
				<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
AI ethical reflection	0.52	728	0.60	5.35	0.65	5.33	0.73
AI ethical awareness	2.59	549.90	0.01	5.78	0.75	5.62	0.86
AI critical evaluation	-1.02	728	0.31	5.20	0.81	5.27	0.87
AI for social good	-0.86	728	0.39	4.84	1.08	4.91	1.13

Finally, we evaluated the convergent validity of the AIERS by investigating the relationship between the AIERS and AI literacy, using the overall scale and its dimensions and conducting a Pearson's correlation analysis (see Table 4). The overall scores for AI ethical reflection and AI literacy were computed by averaging the participants' responses across all of the items for each measure. Using these composite scores, we found a significant positive correlation between the overarching AI ethical reflection factor and AI literacy ($r = .43$, $p < .001$). Moreover, AI literacy showed significant correlations with the three dimensions of the scale, including AI ethical awareness ($r = .19$, $p < .001$), AI critical evaluation ($r = .44$, $p < .001$), and AI for social good ($r = .39$, $p < .001$). These weak to moderate associations provided evidence of both convergent and discriminant validity, suggesting that although AI ethical reflection and AI literacy are related constructs, they remain conceptually distinct. In particular, the relatively lower correlation of AI ethical awareness ($r = .19$) indicated that knowledge of AI does not necessarily correspond to higher ethical awareness in AI use. These results validated the AIERS as a measure that captures a unique aspect of AI understanding, namely AI ethical reflection, which is complementary to yet distinct from AI literacy.

Differences in AI ethical reflection

To better understand AI ethical reflection, the current study further evaluated whether the overall factor, as well as its three dimensions, AI ethical awareness, AI critical evaluation, and AI for social good, differed across students based on their gender, academic discipline (STEM vs. non-STEM), and prior experiences in using AI for learning. The results of independent *t*-tests (see Table 5) revealed no significant differences by gender for the overall AI ethical reflection score or for the subcomponents of AI critical evaluation and AI for social good. However, a significant difference existed for AI ethical awareness, with female students ($M = 5.78$, $SD = 0.75$) reporting higher scores than male students ($M = 5.62$, $SD = 0.86$), $t = 2.59$, $p = .01$. Regarding academic backgrounds, independent *t*-tests (see Table 6) indicated no significant differences between students from STEM and non-STEM disciplines in the overarching AI ethical reflection factor or its

Table 6 Results of independent T-Tests by academic discipline

	<i>t</i>	<i>df</i>	<i>p</i>	STEM students		Non-STEM students	
				<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
AI ethical reflection	0.52	728	0.60	5.33	0.69	5.35	0.68
AI ethical awareness	0.81	683.25	0.42	5.69	0.84	5.74	0.75
AI critical evaluation	0.16	728	0.87	5.22	0.83	5.23	0.84
AI for social good	0.15	728	0.88	4.86	1.10	4.87	1.10

Table 7 Results of ANOVA by AI experiences

	<i>F</i> (2, 727)	<i>p</i>	Never		Less frequent		Frequent	
			<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
AI ethical reflection	18.28	< 0.001	5.19 ^a	0.65	5.26 ^b	0.70	5.58 ^{a, b}	0.61
AI ethical awareness	5.19	< 0.01	5.82	0.77	5.65 ^a	0.83	5.85 ^a	0.71
AI critical evaluation	14.10	< 0.001	5.07 ^a	0.82	5.14 ^b	0.85	5.49 ^{a, b}	0.75
AI for social good	23.69	< 0.001	4.32 ^{a, b}	1.28	4.77 ^{b, c}	1.05	5.26 ^{a, c}	1.04

Note. Means in a row sharing subscripts are significantly different from one another

three dimensions. For AI experience, significant differences were observed for both the overall AI ethical reflection and the three subcomponents based on the students' prior engagement with AI. The results of an ANOVA (see Table 7) suggested that students who frequently used AI tools for learning and assessment in the last term showed higher AI ethical reflection scores than those who reported less frequent use and those who reported never using AI tools. To further identify which groups differed significantly, we conducted Tukey's honestly significant difference (HSD) post hoc tests. For the composite score, the students with frequent AI use had higher overall AI ethical reflection ($M = 5.58$, $SD = 0.61$) than those with less frequent use ($M = 5.26$, $SD = 0.70$) and those who had never used AI ($M = 5.19$, $SD = 0.65$), $F(2, 727) = 18.28$, $p < .001$. The students who frequently used AI reported a higher level of AI ethical awareness ($M = 5.85$, $SD = 0.71$) than those with less frequent usage ($M = 5.65$, $SD = 0.83$), $F(6, 732) = 5.19$, $p < .01$. Frequent AI users also scored higher on AI critical evaluation ($M = 5.49$, $SD = 0.75$) than those with less frequent AI use ($M = 5.14$, $SD = 0.85$) and those who never used AI ($M = 5.07$, $SD = 0.82$), $F(2, 727) = 14.10$, $p < .001$. Lastly, the students who frequently used AI reported the highest level of AI for social good ($M = 5.26$, $SD = 1.04$), followed by those with less frequent use ($M = 4.77$, $SD = 1.05$) and those who had never used AI ($M = 4.32$, $SD = 1.28$), $F(2, 727) = 23.69$, $p < .001$.

Discussion

The present study proposed a three-dimensional model to conceptualize AI ethical reflection in terms of three subcomponents: AI ethical awareness, AI critical evaluation, and AI for social good. The CFA results demonstrated strong evidence of both reliability and validity, suggesting that the AIERS can be a valuable instrument for future empirical investigations and education on ethical reflection in the context of AI. Moreover, significant differences in AI ethical reflection were observed based on the students' gender and prior AI experience, which can provide insights to inform more targeted curriculums to address different characteristics of diverse student populations.

Grounded in Rest's (1986) model of morality and drawing on the two facets of ethical thinking articulated by Goldsmith et al. (2020), the current study regarded AI ethical

reflection as a multidimensional construct that encompasses individuals' awareness of AI ethics, the processes of moral judgment, and motivation/character. However, as both Rest (1986) and Goldsmith et al. (2020) emphasized the fundamental effect of ethical awareness as the initial step in moral reasoning, existing studies have predominantly focused on this domain to explore individuals' competence in recognizing ethical considerations and have further identified its potential subcomponents, such as transparency, fairness, privacy, responsibility, and non-maleficence (e.g., Jang et al., 2022). This narrow focus overlooks the critical roles of evaluative judgment and felt social responsibility in shaping individuals' AI ethical reflection. This study developed a more comprehensive framework that incorporates the full spectrum of moral functioning in the context of AI. We integrated AI ethical awareness with two additional factors, AI critical evaluation and AI for social good, as key factors that comprise AI ethical reflection. AI critical evaluation was contextualized to reflect individuals' critical judgment of AI systems and their outputs, and AI for social good captured individuals' sense of ethical responsibility to use AI in ways that benefit society.

The CFA results supported the three-dimensional structure of the AIERS, validating the three distinct but related dimensions underlying the overarching factor of AI ethical reflection. The findings also revealed a progressive decrease in the mean scores across the three dimensions of AI ethical reflection. Specifically, the participants scored highest in AI ethical awareness, followed by AI critical evaluation, and lowest in AI for social good. Thus, university students may be relatively more capable of recognizing ethical considerations related to AI, whereas they may be comparatively less likely to critically analyze the AI outputs and even less inclined to exhibit behavioral tendencies to use AI for social good. This pattern echoes Rest's (1986) model, which hypothesizes that ethical awareness is the initial step in ethical thinking involving essential recognition of ethical considerations, whereas ethical judgment and motivation/character require deeper cognitive reflection and practical action. In particular, Chai et al. (2020) found that individuals' tendency to use AI for social good tended to be less immediate and influenced by factors such as perceived usefulness and subjective norms. These findings suggest the need for more targeted ethical education that emphasizes the critical evaluations and societal implications of AI to equip students with a more comprehensive ability to reflect on the responsible and ethical use of AI.

Moreover, to further establish the construct validity of the AIERS, the current study investigated the relationship between AI ethical reflection and AI literacy. The results indicated that AI ethical reflection was significantly and positively correlated with AI literacy, while the coefficients were low to moderate in magnitude, suggesting that AI ethical reflection and AI literacy are related yet distinct constructs. This finding aligns with the literature, indicating that ethical reflection can share conceptually interconnected facets with AI literacy (Chiu et al., 2024a; Ng et al., 2021). The students with higher AI literacy, characterized by a deeper understanding of AI and its implications, tended to be more capable of engaging in ethical reflection on the usage of AI with higher levels of abilities for ethical considerations and critical assessment of AI, as well as to have stronger behavioral tendencies to apply AI for social good. However, the low to moderate correlations also offered evidence of discriminant validity, which indicates that AI ethical reflection captures a unique focus on AI ethical awareness, evaluation, and responsibility beyond the knowledge-centered dimension of AI literacy. In light of this finding, further

integration of ethical education into AI literacy curriculums for university students is warranted as a critical way to promote their ethical and reflective engagement with AI. By emphasizing both the technological and ethical aspects of AI, students may develop a more comprehensive understanding of AI, thereby fostering their more responsible use of AI.

This study also investigated whether AI ethical reflection differed based on gender, academic discipline, and prior experiences with AI for learning. Regarding gender, a significant difference was found in the dimension of AI ethical awareness, with female students reporting higher scores than male students. This is consistent with the finding of Jang et al. (2022) that female students demonstrated greater AI ethical awareness. This difference may be understood within the broader context of gender and morality. Previous research has consistently documented significant gender differences in moral sensitivity, which can be attributed to the gender-related socialization of specific roles between females and males (You et al., 2011). Gilligan's (1982) theory of moral development argued that females are more likely to be socialized to prioritize care and relationships, which may give them enhanced sensitivity to ethical issues that impact people. In the context of AI, this may be demonstrated by a greater sensitivity to the relational and societal implications of AI, which may explain the significantly higher levels of AI ethical awareness among female students found in this study. This finding holds important implications for AI education. The observed gender difference in AI ethical awareness emphasizes the necessity of considering the critical role of gender when designing the AI ethics curriculum. AI education programs are suggested to incorporate strategies that cultivate students' moral sensitivity from multiple perspectives of gender to ensure that both male and female students are equipped to recognize the ethical challenges related to AI.

In terms of academic background, the current results showed no significant differences in the overall AI ethical reflection factor or its three subcomponents between STEM and non-STEM students. This finding may suggest that students' ethical reflection on AI is not inherently related to their academic background, highlighting the importance of integrating AI ethics education across all fields. However, AI ethical education has largely been limited to STEM disciplines (e.g., Borenstein & Howard, 2021). Given the widespread presence and far-reaching impacts of AI beyond technical domains, students from all backgrounds should develop the necessary competencies to reflect on ethical considerations related to AI to ensure the ethical and responsible use of AI.

In addition, significant differences in AI ethical reflection were identified based on prior AI experience. Compared with students who used AI less frequently or not at all for learning and assessment in their last term, those who reported frequent use of AI presented significantly higher levels of AI ethical reflection across all dimensions. This finding implies that greater engagement with AI tools may foster a deeper understanding of their ethical implications. This is in line with experiential learning theory (Kolb et al., 2001), which posits that direct experiences can be crucial sources of reflection during the learning process. Accordingly, individuals' direct interactions with AI may enhance their ability to critically reflect on its use. Therefore, incorporating practical AI experiences into AI education can be critical and is expected to further promote students' ethical reflection.

Limitations and future directions

This study has several limitations. First, although the sample had 730 participants, which was large enough for factor analysis, the generalizability of the findings may be limited, as the sample was recruited exclusively from university students in a specific Chinese cultural context. Future studies could seek to validate the AIERS across different populations and settings to confirm its reliability in evaluating individuals' ethical reflection in the context of AI. Second, although the AIERS demonstrated measurement invariance at the factor loading and structural covariance levels, this study found non-invariance at the measurement residual level across genders. According to Byrne (2010), strict invariance at the measurement error level is rarely achieved, and it can be an overly restrictive test for assessing factorial invariance. However, to confirm and explain this non-invariance, future research could explore the potential sources of measurement error that may vary across genders. Additionally, although gender differences in AI ethical awareness were observed in the current study, the underlying factors contributing to these differences were considered based on the existing literature, without further empirical investigation. Future research could empirically explore the role of social factors on gender differences in AI ethical awareness. Furthermore, the present study primarily focused on the development and validity of the AIERS. The potential antecedents and consequences of AI ethical reflection remain largely underexplored. Further studies are needed to examine what factors may influence individuals' ethical reflection on AI and how such reflection may influence individuals' behaviors in their engagement with AI.

Conclusion

The development of the AIERS contributes to the growing body of literature on AI ethics by offering a valid measure of ethical reflection in the context of AI. The three dimensions of the AIERS, namely AI ethical awareness, AI critical evaluation, and AI for social good, provide a comprehensive framework for assessing how individuals reflect on the ethical implications of AI, including recognition of AI's ethical issues, critical evaluation of AI outputs, and commitment to leveraging AI in a socially beneficial manner. Furthermore, the current study identified a positive correlation between AI literacy and AI ethical reflection as well as significant differences in AI ethical reflection based on gender and prior AI experiences. These findings have practical implications for AI education and underscore the need to integrate hands-on AI experiences and ethical education from diverse perspectives into AI curriculums. By promoting students' ethical considerations of AI, educators can cultivate more responsible AI users for the benefit of all.

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Authors' contributions

ZYW: conceptualization, methodology, formal analysis, writing-original draft, writing-review and editing; CSC: conceptualization, methodology, writing-review and editing, supervision, project administration; JIL: conceptualization, methodology, investigation, data curation, project administration; VVYL: conceptualization, methodology, writing-review and editing, supervision, funding acquisition, project administration.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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